



Aluminum Hydroxide Adjuvant

1 Nonproprietary Names

PhEur: Aluminium Hydroxide, Hydrated, for Adsorption

2 Synonyms

Alhydrogel; aluminii hydroxidum hydricum ad adsorptionem; aluminium hydroxide adjuvant; aluminium oxyhydroxide; poorly crystalline boehmite; pseudoboehmite; *Rehydrigel*.

3 Chemical Name and CAS Registry Number

Aluminum oxyhydroxide [21645-51-2]

4 Empirical Formula and Molecular Weight

AlO(OH) 59.99

5 Structural Formula

Structural hydroxyl groups form hydrogen bonds between AlO(OH) octahedral sheets, where hydroxyl groups are exposed at the surface. The surface hydroxyl groups produce a pH-dependent surface charge by accepting a proton to produce a positive site, or donating a proton to produce a negative site. The pH-dependent surface charge is characterized by the point of zero charge, which is equivalent to the isoelectric point in protein chemistry. The surface hydroxyl groups may also undergo ligand exchange with fluoride, phosphate, carbonate, sulfate, or borate anions.

6 Functional Category

Adsorbent; vaccine adjuvant.

7 Applications in Pharmaceutical Formulation or Technology

Aluminum hydroxide adjuvant is used in parenteral human and veterinary vaccines.⁽¹⁾ It activates Th2 immune responses, including IgG and IgE antibody responses. It is also used for the isolation of certain serum components such as blood clotting factors.⁽²⁾

8 Description

Aluminum hydroxide adjuvant is a white hydrogel that sediments slowly and forms a clear supernatant.

9 Pharmacopeial Specifications

See Table I. Note that the USP 32 includes a monograph for aluminum hydroxide gel, which is a form of aluminum hydroxide that is used as an antacid, in which there is a partial substitution of carbonate for hydroxide.

See Section 17.

Table I: Pharmacopeial specifications for aluminum hydroxide adjuvant.

Test	PhEur 6.1
Identification	+
Characters	+
Solution	+
pH	5.5–8.5
Adsorption power	+
Sedimentation	+
Chlorides	≤ 0.33%
Nitrates	≤ 100 ppm
Sulfates	≤ 0.5%
Ammonium	≤ 50 ppm
Arsenic	≤ 1 ppm
Iron	≤ 5 ppm
Heavy metals	≤ 20 ppm
Bacterial endotoxins	+
Assay	90.0–110.0%

10 Typical Properties

Acidity/alkalinity pH = 5.5–8.5

Particle size distribution Primary particles are fibrous with average dimensions of $4.5 \times 2.2 \times 10$ nm. The primary particles form aggregates of 1–10 μ m.

Point of zero charge pH = 11.4

Protein binding capacity >0.5 mg BSA/mg equivalent Al_2O_3

Solubility Soluble in alkali hydroxides and mineral acids. Heat may be required to dissolve the aluminum hydroxide adjuvant.

Specific surface area 500 m^2/g .⁽³⁾

X-ray diffractogram Exhibits characteristic x-ray diffraction pattern having diffraction bands at 6.46, 3.18, 2.35, 1.86, 1.44 and 1.31 Å.

11 Stability and Storage Conditions

Aluminum hydroxide adjuvant is stable for at least 2 years when stored at 4–30°C in well-sealed inert containers. It must not be allowed to freeze as the hydrated colloid structure will be irreversibly damaged.

12 Incompatibilities

When exposed to phosphate, carbonate, sulfate, or borate anions, the point of zero charge for aluminum hydroxide adjuvant decreases.

13 Method of Manufacture

Aluminum hydroxide adjuvant is prepared by the precipitation of a soluble aluminum salt by an alkali hydroxide, or the precipitation of an alkali aluminate by acid.

14 Safety

Aluminum hydroxide adjuvant is intended for use in parenteral vaccines and is generally regarded as nontoxic. It may cause mild irritation, dryness, and dermatitis on skin contact. On eye contact, aluminum hydroxide adjuvant may also cause redness, conjunctivitis, and short-term mild irritation. Ingestion of large amounts may cause gastrointestinal irritation with nausea, vomiting, and constipation. Inhalation of the dried product may cause respiratory irritation and cough. Type I hypersensitivity reactions following parenteral administration have been reported.⁽⁴⁾

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of material handled. Eye protection and gloves are recommended.

16 Regulatory Status

GRAS listed. Accepted for use in human and veterinary parenteral vaccines in Europe and the USA. The limits for use in human vaccines are 0.85 mg aluminum/dose (FDA) and 1.25 mg aluminum/dose (WHO). There are no established limits for use in veterinary vaccines. Reported in the EPA TSCA Inventory.

17 Related Substances

Aluminum phosphate adjuvant.

18 Comments

Different grades of aluminum hydroxide adjuvant with various concentrations, protein binding capacities, and points of zero charge are available.

The impurity limits at 2% equivalent Al_2O_3 are $\text{Cl} < 0.5\%$; $\text{SO}_4 < 0.5\%$; $\text{PO}_4 < 0.1\%$; $\text{NO}_3 < 0.1\%$; $\text{NH}_4 < 0.1\%$; $\text{Fe} < 20 \text{ ppm}$; $\text{As} < 0.6 \text{ ppm}$; and heavy metals $< 20 \text{ ppm}$.

The aluminum hydroxide gel referred to in the USP 32 is used in cosmetics as an emollient, filler, humectant, a mild astringent, and viscosity controlling agent. In pharmaceutical preparations it is used as an adsorbent, and as a protein binder.⁽⁵⁾ It is also used therapeutically as an antacid, and as an abrasive in dentifrices. It is not, however, used as a vaccine adjuvant.

19 Specific References

- 1 Shirodkar S *et al.* Aluminum compounds used as adjuvants in vaccines. *Pharm Res* 1990; 7: 1282–1288.

- 2 Prowse CV *et al.* Changes in factor VIII complex activities during the production of a clinical intermediate purity factor VIII concentrate. *Thromb Haemost* 1981; 46: 597–601.
- 3 Johnston CT *et al.* Measuring the surface area of aluminum hydroxide adjuvant. *J Pharm Sci* 2002; 91: 1702–1706.
- 4 Goldenthal KL *et al.* Safety evaluation of vaccine adjuvants. *AIDS Res Hum Retroviruses* 1993; 9(Suppl. 1): S47–S51.
- 5 Ash M, Ash I. *Handbook of Pharmaceutical Additives*, 3rd edn. Endicott, NY: Synapse Information Resources, 2007; 446.

20 General References

- Gupta RK *et al.* Adjuvant properties of aluminum and calcium compounds. Powell MF, Newman MJ, eds. *Vaccine Design*. New York: Plenum, 1995; 229–248.
- Hem SL, Hogenesch H. Aluminum-containing adjuvants: properties, formulation, and use. Singh M, ed. *Vaccine Adjuvants and Delivery Systems*. New York: Wiley, 2007; 81–114.
- Lindblad EB. Aluminum adjuvants – in retrospect and prospect. *Vaccine* 2004; 22: 3658–3668.
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- Vogel FR, Hem SL. Immunogenic adjuvants. Plotkin SA *et al.*, ed. *Vaccines*, 5th edn. New York: W.B. Saunders, 2008; 59–71.
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21 Authors

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22 Date of Revision

19 February 2009.

Aluminum Monostearate

1 Nonproprietary Names

JP: Aluminum Monostearate

USP-NF: Aluminum Monostearate

2 Synonyms

Aluminum stearate; aluminum, dihydroxy (octadecanoate-O-); dihydroxyaluminum monostearate; *HyQual*; octadecanoic acid aluminum salt; stearic acid aluminum salt; stearic acid aluminum dihydroxide salt; *Synpro*.

3 Chemical Name and CAS Registry Number

Aluminum monostearate [7047-84-9]

4 Empirical Formula and Molecular Weight

$\text{C}_{18}\text{H}_{37}\text{AlO}_4$ 344.50

5 Structural Formula

$[\text{CH}_3(\text{CH}_2)_{16}\text{COO}]\text{Al}(\text{OH})_2$

6 Functional Category

Emollient; emulsion stabilizer; gelling agent; opacifier; stabilizing agent.

7 Applications in Pharmaceutical Formulation or Technology

Aluminum monostearate is mainly used in microencapsulation⁽¹⁻³⁾ and in the manufacture of ointments. Aluminum monostearate is used as a viscosity-increasing agent in nonaqueous cosmetic and pharmaceutical formulations. In addition, aluminum monostearate can be used as an emulsion stabilizer in cosmetic emulsions and is used in cosmetics such as mascara, moisturizers, and sunscreens.

8 Description

Aluminum monostearate is an aluminum compound of stearic acid and palmitic acid. The USP32–NF27 states that aluminum monostearate contains the equivalent of not less than 14.5% and not more than 16.5% of Al_2O_3 , calculated on the dried basis. The JP XV states that it contains not less than 7.2% and not more than 8.9% of aluminium.